

# Verification

## o4\_H\_Simulator\_FVM\_1D\_Compressible\_Heterogeneous

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July 29, 2026

### 1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solution for one-dimensional slightly compressible single-phase flow in a heterogeneous porous medium.

### 2 Governing Equation

For one-dimensional slightly compressible single-phase flow in a heterogeneous porous medium, conservation of mass combined with Darcy's law gives

$$\frac{\partial(\phi\rho)}{\partial t} = \frac{\partial}{\partial x} \left( \frac{\rho k(x)}{\mu} \frac{\partial P}{\partial x} \right), \quad (1)$$

where

- $P$  is pressure [Pa],
- $\rho$  is fluid density [kg/m<sup>3</sup>],
- $\phi$  is porosity [-],
- $k(x)$  is permeability [m<sup>2</sup>],
- $\mu$  is fluid viscosity [Pa·s].

For a slightly compressible fluid,

$$c_f = \frac{1}{\rho} \frac{\partial \rho}{\partial P}, \quad (2)$$

and assuming constant porosity and compressibility, the density is given by

$$\rho = \rho_0 e^{c_f(P-P_0)}, \quad (3)$$

where  $\rho_0$  is the reference density at pressure  $P_0$ . For small pressure variations, the density may be linearized as

$$\rho \approx \rho_0 [1 + c_f(P - P_0)], \quad (4)$$

which leads to the pressure diffusivity equation,

$$\phi c_f \frac{\partial P}{\partial t} = \frac{\partial}{\partial x} \left( \frac{k(x)}{\mu} \frac{\partial P}{\partial x} \right). \quad (5)$$

The domain is defined on

$$0 \leq x \leq L_x, \quad (6)$$

with fixed-pressure boundary conditions

$$P(0, t) = P_L, \quad P(L_x, t) = P_R, \quad (7)$$

and the initial condition

$$P(x, 0) = P_i. \quad (8)$$

### 3 Analytical Solution

To validate the numerical simulator, a steady-state solution is used. In steady state, the flux is constant such that,

$$\frac{d}{dx} \left( k \frac{dP}{dx} \right) = 0. \quad (9)$$

As the flux is constant everywhere, it follows that the Darcy fluxes of the two sections of permeability are equivalent,

$$q = -\frac{k}{\mu} \frac{dP}{dx}. \quad (10)$$

The pressure gradients are therefore different and equal to

$$\frac{dP}{dx} = -\frac{\mu q}{k} \quad (11)$$

in each section. In the first layer, therefore,

$$P(x < a) = P_L - \frac{\mu q}{k_1} x \quad (12)$$

and in the second layer,

$$P(x > a) = P(a) - \frac{\mu q}{k_2} (x - a) \quad (13)$$

where  $a$  is the permeability change location with,

$$P(a) = P_L - \frac{\mu q}{k_1} a. \quad (14)$$

Then, solving for  $q$  yields,

$$q = \frac{P_L - P_R}{\mu \left( \frac{a}{k_1} + \frac{L_x - a}{k_2} \right)}. \quad (15)$$

By substituting in  $q$ ,

$$P(x) = \begin{cases} P_L - \frac{P_L - P_R}{\frac{a}{k_1} + \frac{L_x - a}{k_2}} \frac{x}{k_1}, & 0 \leq x \leq a, \\ P_L - \frac{P_L - P_R}{\frac{a}{k_1} + \frac{L_x - a}{k_2}} \left( \frac{a}{k_1} + \frac{x - a}{k_2} \right), & a \leq x \leq L_x. \end{cases} \quad (16)$$

### 4 Simulation Parameters

The simulation parameters are given in Table 1.

| Parameter                 | Symbol   | Value                              |
|---------------------------|----------|------------------------------------|
| Number of cells           | $N_x$    | 10000                              |
| Domain length             | $L_x$    | 10 m                               |
| First permeability domain | $a$      | 2 m                                |
| Permeability              | $k_1$    | $1 \times 10^{-13} \text{ m}^2$    |
| Permeability              | $k_2$    | $1 \times 10^{-12} \text{ m}^2$    |
| Viscosity                 | $\mu$    | 0.1 Pa·s                           |
| Left pressure             | $P_L$    | $10 \times 10^6 \text{ Pa}$        |
| Right pressure            | $P_R$    | $1 \times 10^5 \text{ Pa}$         |
| Compressibility           | $c_f$    | $5 \times 10^{-8} \text{ Pa}^{-1}$ |
| Reference density         | $\rho_0$ | 1000 kg/m <sup>3</sup>             |
| Reference pressure        | $P_0$    | $1 \times 10^5 \text{ Pa}$         |

Table 1: Simulation parameters.

## 5 Results

### 5.1 Pressure Distribution

The pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, match well.

## 6 Discussion

The numerical pressure distribution reproduces the analytical linear pressure solution at steady state.

## 7 Conclusion

The simulator successfully reproduces the analytical solution for one-dimensional slightly compressible flow in a heterogeneous porous medium. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

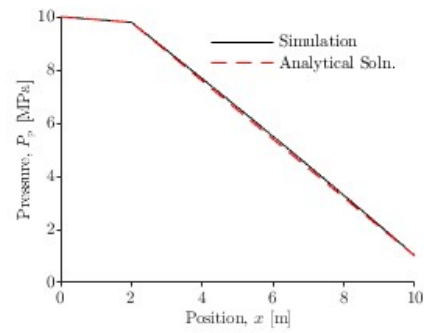


Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator.

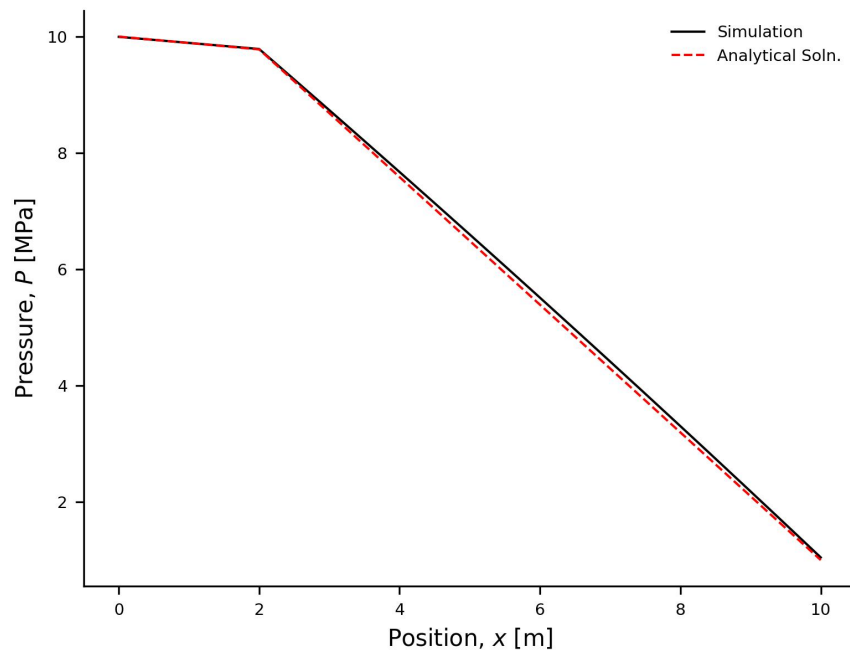


Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator.